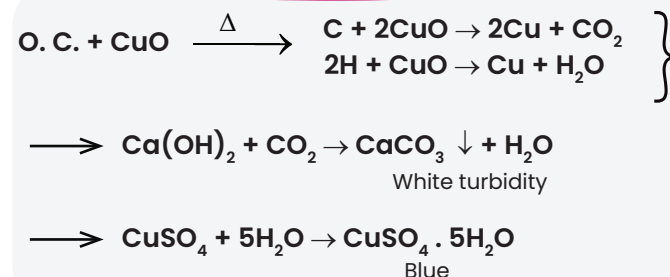
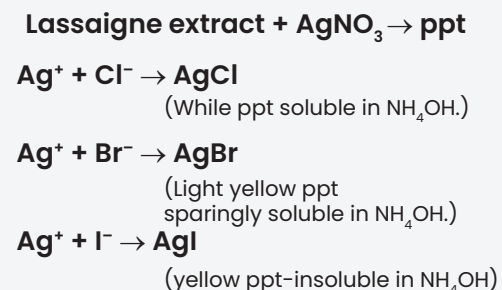


01 Detection of C & H



2a Detection of Halogen:-

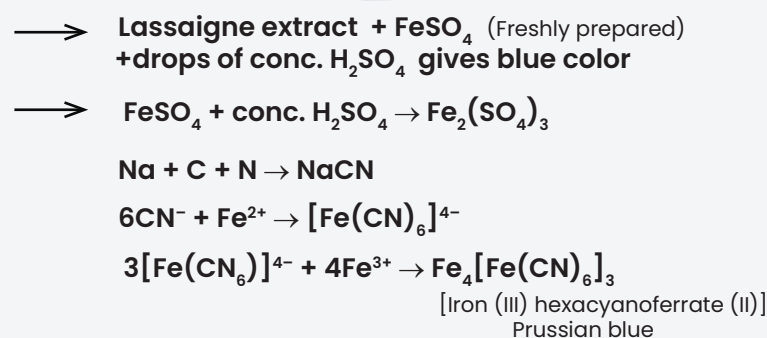


Detection of N, S, P & X

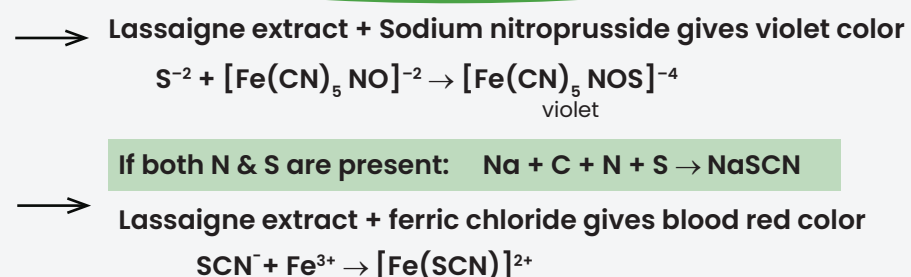
02

Detection of Nitrogen

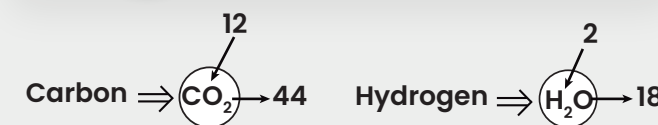
2b



2c Detection of Sulphur



01 Liebig's method



$$\% \text{ of carbon} = \frac{12}{44} \times \frac{M_{\text{CO}_2}}{M_{\text{oc}}} \times 100$$

$$\% \text{ of hydrogen} = \frac{2}{18} \times \frac{M_{\text{H}_2\text{O}}}{M_{\text{oc}}} \times 100$$

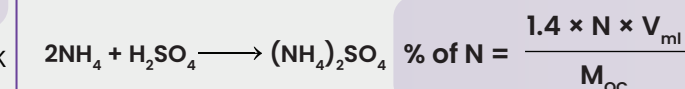
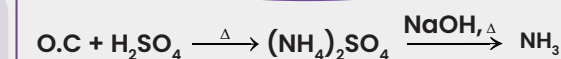
02 Estimation of Nitrogen

(2a) Duma's method

$$\frac{P_1 V_1}{T_1} = \frac{760 \times V_{\text{STP}}}{273} \quad \%(\text{N}) = \frac{V_{\text{STP}}}{22400} \times \frac{28}{m_{\text{oc}}} \times 100$$

Volume of nitrogen collected = V_1 mL, Temperature = T_1 K
 p_1 = Atmospheric pressure – Aqueous tension

(2b) Kjeldahl method



Kjeldahl's method is not used for

• Nitro

• Azo compounds



03 Estimation of X, S, P

Carius method

$$\% \text{ X} = \frac{M_x \times W_{\text{AgX}} \times 100}{M_{\text{Ag}} \times W_{\text{oc}}}$$

$$\% \text{ S} = \frac{32 \times W_{\text{BaSO}_4} \times 100}{233 \times W_{\text{oc}}}$$

$$\% \text{ P} = \frac{31 \times W_{\text{APM}} \times 100}{1877 \times W_{\text{oc}}}$$

$$\% \text{ P} = \frac{62 \times W_{\text{Mg}_2\text{P}_2\text{O}_7} \times 100}{222 \times W_{\text{oc}}}$$

Q. During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.5 g of the compound in Kjeldahl's estimation of nitrogen neutralized 10 mL of $1\text{M H}_2\text{SO}_4$. The percentage of nitrogen in the compound is :

(a) 56%

(c) 50%

(b) 45%

(d) 40%

Q. On complete combustion, 0.246 g of an O.C gave 0.198 g of carbon dioxide & 0.1014 g of water. Determine % composition of carbon & hydrogen in the compound.

(a) 4.58, 21.95

(c) 45.8, 21.95

(b) 21.95, 4.58

(d) 2.195, 45.8

Q. In Dumas' method for estimation of nitrogen, 0.3g of an organic compound gave 50mL of nitrogen collected at 300 K temperature and 715 mm pressure. Calculate the percentage composition of nitrogen in the compound. (Aqueous tension at 300 K = 15 mm)

(a) 17.5%

(c) 6.25%

(b) 28%

(d) 31%

Q. In Carius method of estimation of halogen, 0.15 g of an organic compound gave 0.12 g of AgBr. What is the percentage of bromine in the compound?

(a) 68.08%

(c) 42.1%

(b) 45%

(d) 50%

Q. Which of the following compounds will be suitable for Kjeldahl's method for nitrogen estimation?

